Title: Quiz-Based Approach to Lifestyle Behaviour Change Using LLMs

- Should LLM be Large Language Model in the title (instead of an abbreviation).

Strengths/Contributions:

- Strong introduction and background with thorough explanation of the problem, the plan and LLM use.
- This seems like a comprehensive system design (complete pipeline).
- The weightings throughout for different aspects are a nice inclusion and detail some of the thought processes.

Presentation/Structure/Language:

- Is there a plan for an Abstract?
- Presentation is clean and well organised.
- Some areas could be shorter space wise (notably section 2.4)
- The methods section can be difficult to follow at times due to the complexity of some of the topics (for example section 3.5.1, what are embeddings?)

Reference Use:

- Good reference use regarding the relevant background (obesity, learning behaviour, etc).
- Could include more references, particularly regarding LLM's and possible problems or the use of AI in healthcare.
- More references within the methods section could clarify decisions made.

Figure/Table/Formula Use:

- Figure 2 needs more of a label. Can't have (a) – (f) without descriptions. Same with Figure 3. Both could also be larger for improved clarity. Could move the descriptive paragraphs to be mostly in the label.
- Tables look mostly alright. Table 5 seems to have just one issue and probably doesn't need to be a table.

Approach/Improvements:

- The aim is not appropriate. It states that the focus is on evaluating the effectiveness of an AI learning approach, but the report focuses on the development of such. There is no evaluation of the effectiveness (no tests/subjects etc)
- Could discuss more other existing solutions and their similarities/differences. Consider discussing more alternative approaches.

- More information on the useability evaluations and the participants involved. Was it just the authors? Was it someone externally? How were scores collated for Table 4, are they averaged?

Other Comments:

- Could make it clearer the use of a mobile application in this project.
- Hypothesis seems unnecessary. If you didn't expect this to work you wouldn't do the research. It also states that the LLM approach will be more effective but doesn't do this comparison.
- Consider changing the use of 'We' to more objective language to fit more consistently within the report.
- Could include a more thorough explanation of app development process (GUI, etc).
- Overall, the language throughout the report could be modified to make it clearer for a reader with less of a relevant background than the authors.
- The tense varies throughout the report between past and present (were / are) and this should be made consistent.

Below is the report with markings and comments.

Quiz-Based Approach to Lifestyle Behaviour Change Using LLMs

Health Systems / Computer Science Track

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1 Introduction

Obesity worldwide has increased tenfold over the past four decades [2]. This alarming trend is associated with numerous health risks and substantial societal costs [9]. Diet plays a key role, with increasing consumption of high energy foods and decreasing intake of fruits and vegetables. However, achieving and sustaining healthy dietary behaviors remains challenging.

Dietary counseling is an established approach in which patients work with healthcare professionals to set goals and create personalized plans [3]. However, behavioral change is time consuming and requires significant effort from both patients and professionals [5].

Bälter et al. [1] propose learning solely through questions, using a method called pure question-based learning (pQBL). This approach requires no background content; instead, feedback is provided when a question is answered incorrectly, guiding learning iteratively. The rapid development of AI tools presents opportunities to apply it in health related contexts, such as automatically assessing dietary choices within an app [4]. Which in turn could be tool to automate the generation of questions in pQBL.

1.1 Project Aim

The aim of this project is to evaluate the effectiveness of an AI-generated pQBL learning approach in promoting sustainable dietary behavior change. The method is implemented in a Health app, incorporating:

- Adaptive question-based learning using an LLM,
- Personalized content generation based on health data and dietary input,
- Progress tracking through daily streaks

1.2 Hypothesis

We hypothesize that the pQBL approach supported by LLMs will be more effective than traditional fact-based learning in promoting sustainable dietary improvements.However, we hypothesize that the LLM will struggle with fact checking the questions.

2 Background

2.1 Obesity and Dietary Behavior

Middle aged and older adults are particularly vulnerable to obesity and metabolic disorders due to aging, stress, sedentary lifestyles, and irregular diets [7]. Dietary behavior plays a crucial role in shaping obesity metabolic types, which are classified as metabolically healthy non-obesity (MHNO), metabolically unhealthy non-obesity (MUNO), metabolically healthy obesity (MHO), and metabolically unhealthy obesity (MUO) [7].

Unhealthy habits such as salty, greasy, or spicy foods and fast eating increase the risk of metabolic abnormalities. While healthier behaviors: light or moderate taste, moderately firm food texture, slower eating speed, and daily intake of dietary supplements, are protective. Individuals with lower healthy eating scores (0–2 or 3–4 of 6) were significantly more likely to develop MUNO, MHO, and MUO, while higher scores (5–6) were associated with lower BMI, triglycerides, and fasting glucose, and higher HDL cholesterol. These findings highlight that adopting balanced dietary behaviors is essential for preventing and controlling obesity related metabolic diseases in aging populations.

2.2 Pure Question-Based Learning (pQBL)

Bälter et al. [1] describe several learning approaches, including Question-Based Learning (QBL), content-biased learning, and pQBL. pQBL draws from active learning principles and relies solely on questions and constructive feedback, without presenting any prior learning material. Instead, the educational content is embedded within the feedback provided after an incorrect answer. This contrasts with QBL, where content is introduced before the questions, and feedback is used only to reinforce understanding. In content-biased learning, feedback refers back to the pre-existing course material rather than presenting new information. According to Bälter et al. [1] pQBL reduces the amount of learning materials teachers has to produce and it also breaks down the learning to smaller pieces.

2.3 Use of LLMs in Personalized Learning

Traditional learning techniques have limitations, particularly regarding personalization [8]. LLMs perform well with large datasets, which enables them to leverage personalized information to generate tailored learning content. They can offer real-time adaptation of difficulty, pacing, and feedback. Features that are difficult for a teacher to provide consistently when managing many students. However, Sharma et al. [8] also highlight that LLMs may negatively affect critical thinking and problem-solving abilities. There-

fore, they emphasize the importance of integrating LLMs into learning environments in a balanced and responsible way to support, rather than replace, cognitive development.

2.4 Behavior Change Techniques

There are a large number of behavior change techniques, and Del-Valle-Soto et al. [5] highlight eight commonly used ones: self-monitoring, feedback on behavior, goal setting, social support, rewards and incentives, reminders and alerts, personalization, and education and information provision. In this project, we focused on several of these.

2.4.1 Self-monitoring

The ability for the user to track their own data and performance.

2.4.2 Feedback on behavior

Providing users with feedback about their actions. For example, showing statistics on how well they answer questions.

2.4.3 Goal setting

Allowing users to define goals, such as losing a specific amount of weight or completing a learning milestone.

2.4.4 Rewards and Incentives

Motivating users to return to the application through rewards, badges, or other forms of reinforcement.

2.4.5 Personalization

Tailoring the content and experience based on user input and behavior.

2.4.6 Education and Information provision

Exposing users to relevant educational content inside the application to support behavior change.

3 Methods

3.1 Technical Implementation

- Front-end: Implemented using Natively.
- Data structure: TypeScript-based question bank.

3.2 Application System Design

3.2.1 User Profiling and Data Acquisition

To enable personalized recommendations, the system constructs a multi-dimensional user profile based on self-reported data. The data collection process categorizes information into three distinct layers:

- Demographic Profile: Captures essential biological metrics including Gender, Age, Height, and Weight. Notably, to prioritize user privacy and streamline data entry, quantitative attributes are collected as categorical ranges (e.g., Age '18-25', Weight '60-70') rather than exact numerical values.
- Dietary Behavioral Metrics: Assesses current lifestyle habits, specifically frequency of exercise, sleep duration, meal patterns, vegetable consumption, and sugary beverage intake.
- Health Intentions: Identifies the user's primary goal, acting as the core filter for content. Users must select one of five objectives: Improve eating patterns, Reduce sugar intake, Increase vegetable/fruit intake, Improve protein intake, or Healthy weight management.

3.2.2 Question Bank Architecture

The system is built upon a structured, pQBL-compliant architecture designed to deliver personalized health education. The core of the system is the Question Bank, which utilizes a strict schema to ensure pedagogical consistency and technical scalability. Each question entry in the database is a structured object containing the fields shown in Table 1.

Table 1: Data Structure of the Question

Field Name	Description & Specification
Core Content (Interaction & Feedback)	
`id`	Unique Identifier (UUID) for system traceability).
`question_text`	The main text content of the question.
`answer_options`	An array of potential response strings.
`correct_answer`	Integer (0-based index) indicating the valid response option.
`explanation`	Educational feedback providing immediate behavioral guidance or learning tips.
Classification & Tagging	
`health_intention`	Categorical string matching the user's selected goal. Used as the primary hard filter.
`tags`	A unified array aggregating Content Tags (topics), Behavioral Tags (e.g., `sugar_high`), and Demographic Markers (Age, Gender, Food preferences) for weighted relevance scoring.
`difficulty`	Integer level (1-3) representing cognitive load: 1=Awareness, 2=Reflection, 3=Planning.
System Metadata	
`source`	Origin indicator: `'manual'` or `'LLM-generated'`.
`created_at`	ISO 8601 timestamp recording generation time.
`version`	Batch identifier for content updates and version control.

3.2.3 Personalized Recommendation Algorithm

The core of the system's personalization engine is the Filtering Algorithm. This algorithm is designed to curate a list of dietary and health-related questions that are strictly relevant to the user's unique profile. The process operates through a multi-stage pipeline involving strict filtering, weighted scoring, and randomized selection to ensure high relevance and variety.

Stage 1: Health Intention Filtering (Required) It filters the global database of questions to isolate only those that align with the user's primary dietary goal.

Stage 2: Tag-Based Scoring System Once filtered by intention, the system applies a scoring logic to rank questions based on personal relevance.

a. Profile-to-Tag Conversion: The user's raw profile data is normalized into system tags via the `convertProfileToTags()` function:

- *Age:* `young_adult`, `senior`
- *Lifestyle:* {`sugar, vegetable, exercise, sleep, meals`}`_high/low`
- *Dietary Preferences:* `vegan`, `vegetarian`, `gluten_free`, `dairy_free`

b. Weighted Scoring Logic: The system compares the user's tags against the tags associated with each question. As shown in Table 2, specific health factors are prioritized:

Table 2: Tag-Based Scoring Weights

Tag Category	**Weight**	**Rationale**
Sugar-related	+3	Due to the large number of obese people and the fact that modern diets are typically high in calories, obesity is a high priority
Lifestyle	+2	Moderate priority (Exercise, Sleep, Vegetables)
Bonus Match	+1	Applied if question matches *any* user tag

Stage 3: Sorting & Selection The final stage ensures the delivery of the top-ranked content while maintaining user engagement through variety:

a. Sorting: Questions are ordered by their calculated score in descending order.

b. Randomization: To prevent repetition, the algorithm randomizes the order of questions that share the same score.

c. Selection: The system returns the top 5 questions.

3.3 LLM-Based Question Generation Pipeline

The system implements an automated workflow to generate high-quality health and nutrition questions using OpenAI's `gpt-4o-mini` model. This pipeline orchestrates prompt engineering, API interaction, semantic deduplication, and schema validation to ensure the integrity of the educational content.

3.3.1 System Architecture

The generation logic is built on a Node.js/TypeScript stack, comprising four primary components: the entry script (`start-dev.ps1`) for environment configuration, the orchestrator (`batchGenerateQuestions.ts`) for lifecycle management, the core service layer for API communication, and the existing data store used for reference during deduplication.

3.3.2 Workflow Implementation Details

Stage 1: Configuration and Prompt Engineering The workflow initiates by validating environment variables (API keys and model selection). The core logic then constructs a strictly typed system prompt via the `buildBatchPrompt()` function. Unlike generic prompts, this system enforces a rigid JSON schema output and mandates coverage across five specific health intentions. It explicitly requests a balanced distribution of difficulty levels (1–3) and necessitates the inclusion of specific metadata tags to support the personalized recommendation engine.

Stage 2: Execution and Data Transformation The system executes a batch request to the OpenAI API with a temperature setting of 0.8 to ensure creative diversity in the question text. The token limit is dynamically calculated based on the requested batch size (approx. 350 tokens per question). Upon receiving the raw response, the system strips Markdown artifacts and parses the content into the internal `QuizQuestion` objects. During this transformation, system metadata, including unique UUIDs, timestamps, and version control tags are injected into each record.

Stage 3: Quality Assurance (Deduplication & Validation) To prevent content redundancy, the pipeline employs a Jaccard Similarity algorithm. The system normalizes the text of incoming questions (removing punctuation and casing) and compares them against the entire existing database. Any question sharing a similarity coefficient greater than 0.70 (70%) with an existing entry is automatically discarded. Following deduplication, a validation layer checks structural integrity. It enforces constraints such as minimum text length, exact option counts, and valid integer ranges for correct answers.

Stage 4: Output and Error Handling Verified issues are serialized into timestamped JSON files and stored in the ‘data/generated/‘ directory, awaiting subsequent fact checking. The system implements a robust error handling mechanism specifically for managing API rate limits (HTTP 429), employing an exponential backoff strategy with a maximum of 3 retries.

3.3.3 Process Visualization

Figure 1 illustrates the sequential logic from initialization to storage.

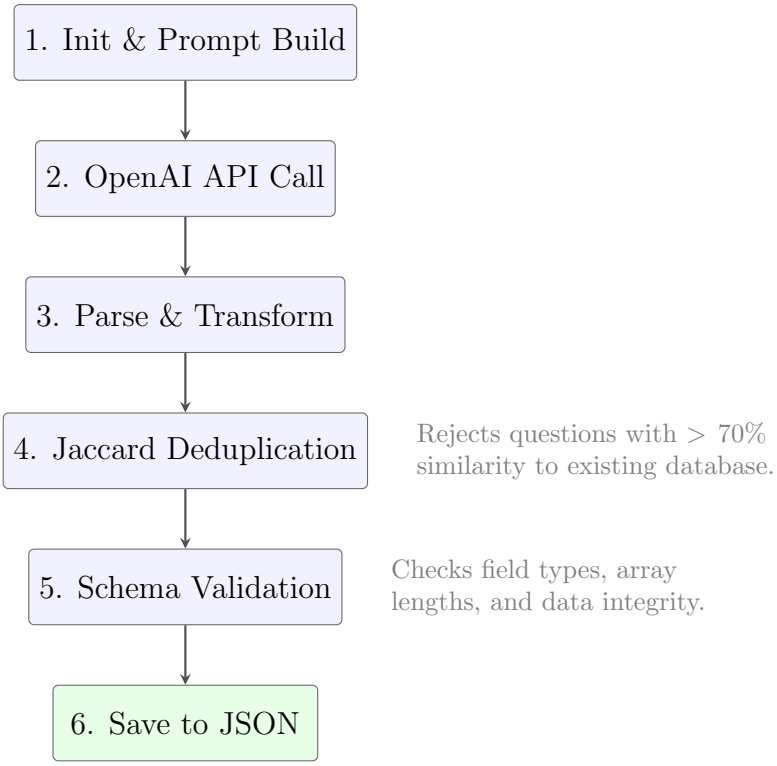


Figure 1: Automated Question Generation Workflow

3.4 LLM Cross Validation

3.4.1 High-level structure

As part of the research made to find supporting evidence for choosing a suitable LLM for the first version of the application, a cross validation was made, evaluating question generating performance in between three different LLM's. The LLM's chosen are four well known LLM's, in three different price ranges with two of them being low-cost models and one in a mid-tier price class. The cheapest one chosen was Gemini 2.5 Flash. Close in range, GPT-4o Mini was chosen, and finally, Claude Sonnet 4.5 being the most expensive one [10].

First, three different question banks were generated through providing each LLM with specific instructions that provide the LLM with explicit information about allowed sources (for this research state, only two sources were used for easier control of output), output format, specific aim of questions generated etc., see appendix 1. A batch of 100 questions were generated by each LLM, saved as bilingual JSON question sets since the final chosen question bank will be used in a manual fact checking system that uses both an English and Swedish version of each question when running the fact checking pipeline (see chapter Manual Fact-Checking).

The question banks are then loaded into a simplified, English-only, *SimpleQuestion* format for evaluation in *crossValidateQuestionBanks*.ts. The simpler format contains id, text, answer options, correct index, explanation to question answer, and health intention. All evaluators (based on OpenAI, Claude, and Gemini) then use the same system prompt and scoring rubric:

- clarity
- correctness (aligning with information provided from limited sources)
- alignment (with the stated health intention)
- pedagogy

Each LLM is instructed to return pure JSON with *model_name*, *evaluated_bank*, *question_score* with per-question percentage scores (0-100), and *overall_score* for the whole question bank evaluated.

3.4.2 LLM implementation

First, the OpenAI evaluator builds a *systemPrompt* describing the role and evidence rules based on the given specific instructions. Then, a *userPrompt* containing a serialized list of *SimpleQuestions* and a specific JSON schema with an example of the expected output is created. The response from the API is then read as plain text and parsed directly as JSON into an internal *EvalResult* structure. Claude and Gemini, reuse the exact same rubric and JSON schema but adapt it for each provider's specific API to enable identical evaluation rubric. For Claude, the script needs to call the messages API with the same prompts. Next, it post-processes the returned text, removing any optional Markdown code fences (“‘) before parsing it as JSON to match the structure for *EvalResult*.Gemini does a similar process, calling Google's generative language API, extracts the first candidate text, strips it from potential code fences, and then parses the result for *EvalResult*.

3.4.3 Cross validation pipeline

After the pipeline loads the three question banks generated by each LLM (chatGPT, Claude and Gemini) from the JSON files and extracts the English-only simplified *SimpleQuestion*,a subset of 50 questions per question bank (for managing amount of API calls) is extracted using the *CROSSVAL_SAMPLE_SIZE* environment variable. The sample questions are the first N selected sequentially from the bank.

Evaluators are executed one at a time, each evaluating all available question banks (including its own) by calling the appropriate *evaluateWithX* function, with the sample size

controlled by the *CROSSVAL_SAMPLE_SIZE* environment variable. The evaluator then saves the raw evaluation results as JSON files containing per-question scores in the same rubric mentioned before. A statistical summary is also generated via *summarizeEval* in the console, providing aggregate metrics.. Correctness score is disaggregated by health intention category, for example "Increase vegetable and fruit intake".

3.5 Manual Semantic fact checking methodology

The manual fact-chacking method includes a semantic fact-checking technique that verifies the questions in the applications question bank against a PostgreSQL database containing the two sources chosen for testing, Harvard Nutrition Source, and *Livsmedelsverket* (The Swedish Food Agency). The system uses semantic embeddings [6] and rule-based keyword matching, eliminating the need of an evaluation with an LLM. The system is described on a high level below. For full specification a markdown file *factChecker.md* is generated.

3.5.1 Embedding and key-word strategy

For each question from the question bank, three embeddings are created, one for the question only, one for the answer only, and then a joint embedding combining both question, answer and explanation. This technique captures semantic signals across components. This system uses OpenAI's *text_embedding_3_small* as default with an abstraction allowing custom embedding functions. The text is truncated to 20 000 characters to stay within limits (8 000 tokens), with a sentence boundary preservation.

Then, a rule-based keyword extractor identifies health related terms from question, answer, and explanation. It uses predefined bilingual keywords in English and Swedish, covering keyword groups like sleep, exercise, food groups, vitamins, minerals, diet etc. The system then scans the question, answer, and explanation and looks for any match with the predefined keywords. In addition to the fixed list, the extractor also looks for individual tokens (keywords) that are three characters long or longer, and appear amongst health terms. All keywords from the three components are then merged into a joint keyword set.

3.5.2 Scoring Algorithm and Relevance Gating

For scoring, the questions are compared to the database in chunks, with the help of each of the three embeddings (question, answer, joint question, answer, and explanation), looking at how well the database matches the contents of the embeddings. These three similarity scores are then blended with preset weights (joint signal counts the most, then question alone, then answer alone). If the chunk of text from the database shares health

keywords with the question/answer, it gives the question a small bonus that grows with more overlap. Then the blended score is normalized between 0-1 and the text chunk that produced the strongest signal is tagged.

The relevance of each chunk is decided through a two stage threshold gate: kept if $finalScore \geq minSimilarity$ with a default at 0.30, or $keyWordOverlap \geq minKeyOverlap$. Remaining chunks are then sorted by their final score in descending order, and the top-K (default : 5) are selected. The top-K scores are then averaged to produce an aggregated score and then converted into a relevance percentage (0–100%). A question is considered supported if $relevancePercentage \geq 75\%$, and gets bonuses for two and three sources (5% boost for three sources, and 2% for two sources).

3.5.3 Bilingual process and Database Integration

To avoid having to build the database on sources only given in one language (English or Swedish based), the fact-checking runs separately for Swedish and English, extracting each question in each language. Then the results are combined by the following:

a. Merging supporting sources and deduplicating by URL, keeping the highest relevance

b. Combining the scores by averaging from each language results (Boost with 20% if both languages are supporting)

c. Recalculating scores based on multi-source bonuses

Text chunks from the database are retrieved from PostgreSQL *data_table*, filtering rows with non-null data and length ≥ 50 characters. Each chunk includes an *ID, text content*, and optional source *URL*. The system supports embedding caching on chunks to reduce API calls during batch processing.

3.5.4 Output format

After running the fact-checker, the system returns a structured *FactCheckResult* containing: question metadata (*ID, text, correct answer, explanation*), binary support status (*isSupported*), relevance percentage (0-100), normalized score (0-1), supporting sources array (with ID, URL, relevance score, excerpt, matched terms, match type), and finally, a rule-based textual summary indicating support level (Strongly supported $\geq 75\%$, Supported 50–74%, Partially supported 30–49%, Weakly supported 1–29%, Not verified 0%).

3.6 Usability Evaluation

3.6.1 Study Design and Participants

To evaluate the system's usability, engagement, and technical stability, a formative usability test was conducted with a small sample size of three. The evaluation utilized a persona-based approach. Instead of inputting their own sensitive health data, participants selected one of five pre-defined user personas (Table 3) to simulate different demographic profiles and health goals. This ensured that the system could be tested while maintaining user privacy.

Table 3: User Personas used for Usability Testing

Persona	**Key Characteristics & Health Goal**
Young Professional	Busy lifestyle, eats processed foods. Goal: *Improve eating patterns.*
Health-Conscious Student	Vegetarian, active. Goal: *Increase vegetable/fruit intake.*
Middle-Aged Parent	Sedentary, high sugar intake. Goal: *Reduce sugar intake.*
Active Retiree	Gluten-free, moderate activity. Goal: *Healthy weight management.*
Fitness Enthusiast	High activity, strict routine. Goal: *Improve protein intake.*

3.6.2 Testing Procedure

The sessions lasted approximately 30–40 minutes and followed a structured protocol consisting of four phases:

1. **Introduction and Consent:** Participants were briefed on the purpose of the study and asked to select a persona.
2. **Free Exploration (3 min):** Users were allowed to explore the interface without specific instructions to assess design.
3. **Structured Task Execution (15–20 min):** Participants were asked to perform six specific tasks designed to cover the core user journey:
    - **Task 1: Onboarding:** Complete the profile setup using persona data.
    - **Task 2: Quiz Execution:** Complete a "Todo" task (5-question quiz) and view the summary.
    - **Task 3: Task Map Navigation:** Interpret the status of tasks (e.g., locked vs. available).
    - **Task 4: Interruption Recovery:** Exit and reopen the app to test state preservation.

- **Task 5: Review Mode:** Access and review previously incorrect answers.
- **Task 6: Profile Management:** View statistics and personalized recommendations.

4. **Post-Test Interview:** A debriefing session to collect qualitative feedback and quantitative ratings.

3.6.3 Data Collection and Metrics

Data was collected using a different methods:

- **Quantitative Metrics:** Participants rated specific features (Onboarding, Quiz System, Interface) and overall satisfaction on a 5-point scale (1 = Poor, 5 = Excellent).
- **Issue Logging:** Technical errors and usability hurdles were logged and categorized by severity (High/Medium/Low). High severity was defined as an issue preventing task completion.
- **Qualitative Feedback:** Open-ended questions regarding the user's favorite and least favorite features.

4 Results

4.1 Prototype Overview

Figure 2 shows the quiz flow. (a) The data collection interface gathers initial user profiles. (b) The home page features a task map comprising six tasks, requiring users to complete at least one task daily. (c) The quiz interface displays question content and formatting, while (d) provides immediate post-response feedback. To enhance engagement, (e) random 'quick tips' appear during quizzes to supplement learning. Finally, (f) offers a performance review upon task completion.

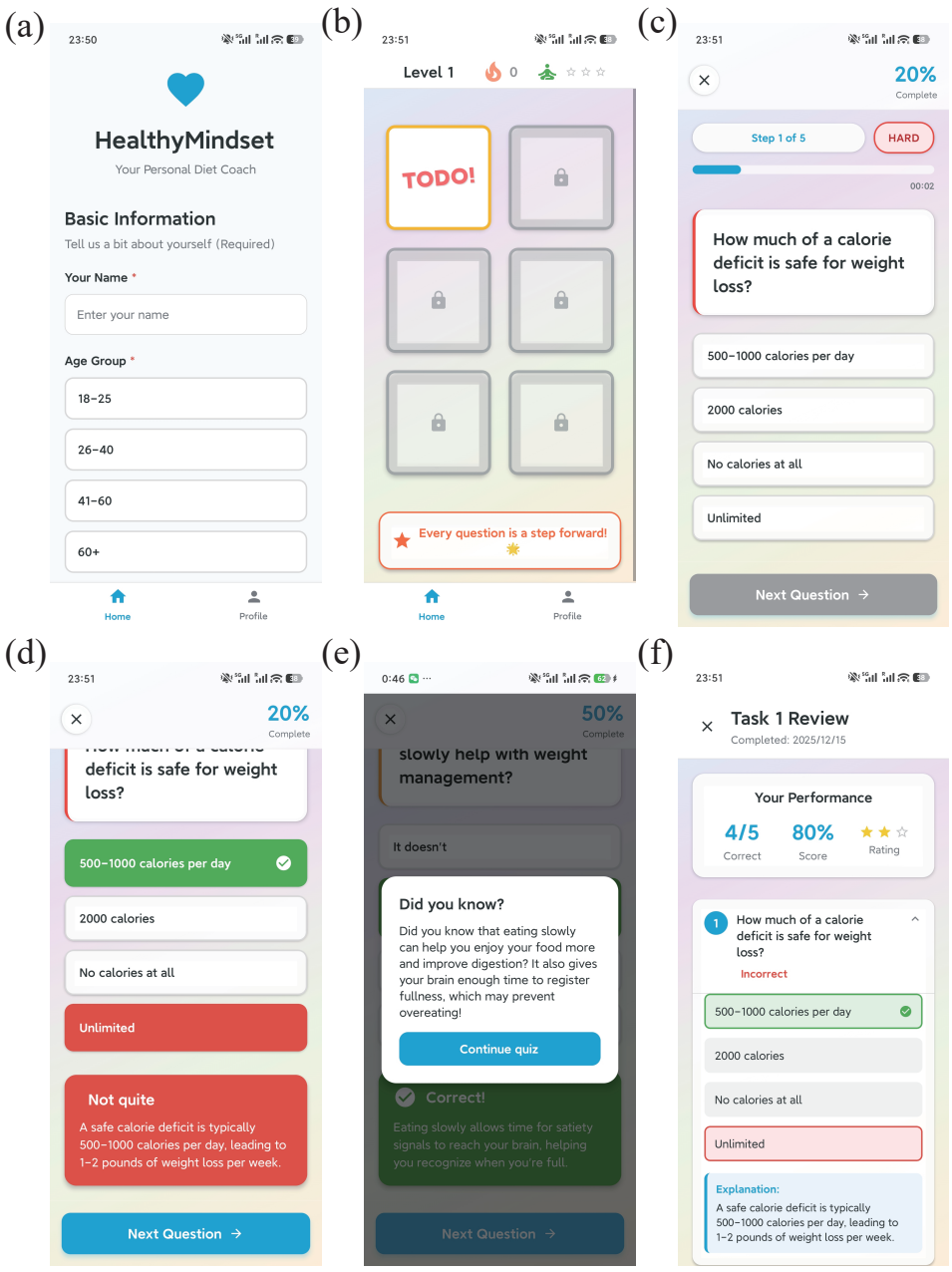


Figure 2: Prototype interface showing quiz flow

Figure 3 shows profile page features. Panel (a) presents key statistics, including user streak and overall accuracy. Selecting the 'View Detailed Analysis' button generates the health report depicted in (b), which visualizes performance trends and offers tailored recommendations aligned with specific health goals.

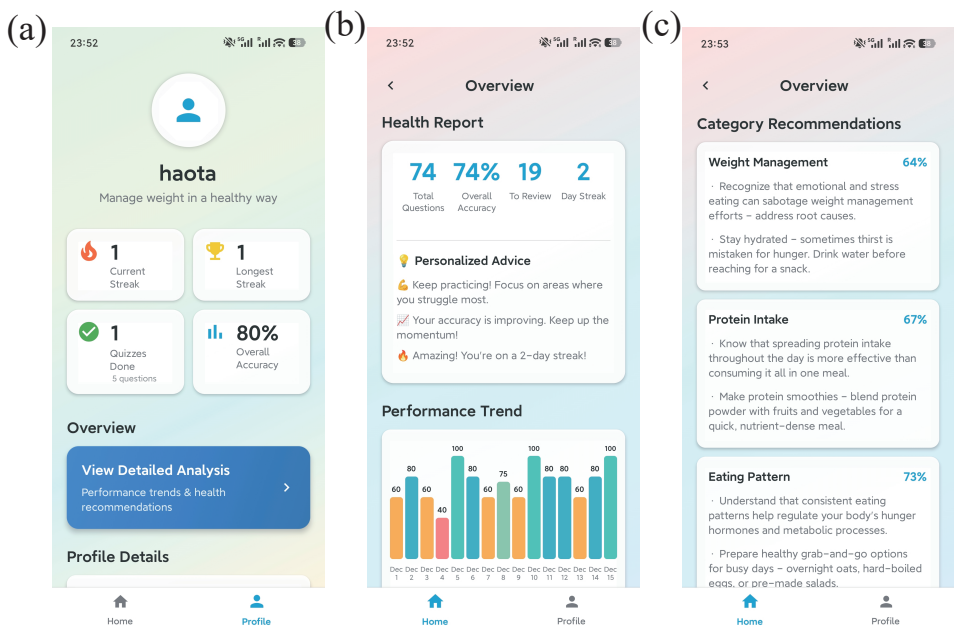


Figure 3: Profile page feaetures

4.2 Evaluation Outcomes

Overview of user responses and observed engagement levels. The following tables summarize the results from usability testing conducted with three personas: the Middle-Aged Parent, the Fitness Enthusiast, and the Young Professional

4.2.1 Overall Consensus Ratings

The application was generally rated as functional but lacking in depth. Onboarding and the Quiz System were the highest-rated features, while Feature Completeness and the Task Map received the lowest scores due to confusion and lack of content.

Table 4: Consensus Metrics and Ratings across all Personas

Metric	Rating	Key Insight
User Onboarding	4/5	Consistently rated as clear, simple, and logically presented across all user types.
Quiz System	4/5	Users found the questions relevant and easy to understand, though the content library is limited.
Overall Satisfaction	3.3/5	Functional but simplistic. Users felt the app lacked the depth required for long-term engagement.
Usability	3/5	The core flow (Q&A) is intuitive, but the Homepage/Task Map navigation is confusing.
Interface Aesthetics	2.7/5	Polarizing results; while one user liked the clean design, the majority found it "too simple" and lacking visual appeal.
Task Map	2/5	Universal confusion regarding the purpose of the "6 TODOs" and the lack of daily structure.
Feature Completeness	1.3/5	The significant lack of features (e.g., gamification, reminders) was the primary detractor for all users.

4.2.2 Critical Issues Log

A single critical severity issue was identified during testing that prevented users from continuing their sessions.

Table 5: High-Priority Issues Identified

Issue Description	Severity	Impact & Root Cause
Infinite Loop / Content Exhaustion	**CRITICAL**	**Impact:** Users are blocked from progressing after completing 3–4 tasks. The app enters an infinite loop or fails to load new questions. **Root Cause:** Content scarcity in the database (specifically "Protein Intake") and a lack of robust error handling for empty search results.

4.2.3 Recommended Improvements

Participants provided consistent feedback focusing on the need for increased engagement mechanisms and clearer workflow structures.

Table 6: Consensus Improvement Suggestions

Category	User Suggestions
Gamification	Users strongly requested gamification elements similar to *Duolingo* or *Candy Crush* (e.g., leveling up, climbing a mountain, streaks) to combat boredom.
Content & Structure	Significantly expand the question library to prevent the infinite loop bug. Curate progressive "learning paths" for each health goal rather than random tasks.
Task Clarity	Replace the ambiguous "6 TODOs" with a clear daily structure (e.g., "Monday", "Tuesday"). Clearly communicate the expected daily workload.
Visuals & Stats	Improve visual aesthetics with more interactive elements. Refine statistics to show meaningful data (Correct vs. Incorrect answers) rather than just "Total Questions."

5 Discussion

5.1 Interpretation of Findings

The usability testing yielded valuable insights regarding user engagement and clearly highlighted areas where the initial prototype required improvement. Participants expressed a consensus that the application lacked depth and required clearer instructions. However, the core quiz mechanics and content were perceived as relevant, with feedback indicating that the pQBL approach was intuitive and effective. As noted by [1], this approach is likely to enhance learning outcomes.

A critical limitation identified during testing was the insufficient volume of questions, which restricted the variability of quizzes for different user personas. Fortunately, this is a straightforward issue to resolve by expanding the question bank. Additionally, while the development phase prioritized simplicity to ensure ease of use, testing revealed that this focus resulted in a lack of engagement. The feedback suggests that while simplicity is essential, the current design was perceived as overly simplistic, failing to maintain user interest.

5.2 LLM Performance

Under construction.

5.3 Comparison with Traditional Methods

Our approach utilized pQBL, as described in the methods section. This technique is new and relatively untested compared to traditional content-based learning, which relies on students passively acquiring knowledge during classes. Bälter et al. [1] identifies this traditional approach as passive learning, arguing that it does not support retention as effectively as active learning. By using pQBL, we aim to provide active learning opportunities where users are pushed to engage with the material directly rather than simply receiving it. This method also makes the learning process more fun and engaging.

5.4 Limitation

A significant limitation was our lack of prior experience in mobile application development. This necessitated a heavy reliance on LLMs for code generation. While this accelerated the initial work, it significantly constrained our freedom and resulted in poor error handling, as the generated codebase was difficult to debug and modify manually. Additionally, strict time constraints limited the project scope, forcing the prioritization of core functionality over a more comprehensive feature set.

Furthermore, the evaluation was confined to a short-term usability study with a small sample size. Consequently, while we could assess immediate user engagement and usability, the study cannot provide empirical evidence of long-term behavioral change or habit retention.

Despite the robust architectural foundation, several limitations persist within the current system design. A primary constraint lies in the reliance on self-reported data for user profiling. Metrics such as weight, vegetable consumption, and exercise frequency are subject to recall bias and social desirability bias, potentially leading to a skewed user profile that does not accurately reflect the user's physiological reality.

5.5 Future Implementations

A priority for future development is the completion and integration of the manual fact-checker, which is essential to ensure the clinical accuracy of AI-generated content. Additionally, the system's personalization capabilities could be significantly enhanced by integrating with wearable health technology. By accessing real-time physiological data, such as sleep duration, the system could dynamically generate context aware educational content. For example, generating questions regarding metabolic consequences of sleep deprivation following a night of insufficient rest.

Furthermore, upgrading the LLM pipeline to support image recognition, would allow for diet analysis based on user-uploaded photos. As demonstrated by Brevegård et al. [4], image recognition technology can identify meal components; integrating this would make the generation of quizzes directly related to the user's specific meal choices. Finally, the engaging nature of the pQBL format suggests potential for cross-domain application. The framework could be adapted to increase interest in complex information in other sectors where user engagement is traditionally low.

Regarding the recommendation engine, the current weighted scoring logic should be replaced or augmented by a Reinforcement Learning (RL) agent. An RL-based approach would allow the system to optimize content delivery based on real-time feedback loops, automatically adjusting the difficulty and topic distribution based on the user's quiz performance and retention rates.

6 Ethical and Sustainability Considerations

6.1 Ethical Issues

6.1.1 Unclear intended use and lack of transparency

If the intended use as well as user experience is not clear, unsuitable users might have a bad experience with the app. For instance, if the instructions for intended use do not clearly state that the app is for educational purposes only and that all information provided is merely a recommendation, the absence of user health data may result in advice that is less tailored and potentially less effective for the user. Also, leaving out warnings and precautions can also lead to a bad user experience and have a negative effect on the users physical and mental health.

Solution: Form a clear instruction of intended use as well as a description of what should be expected by the app. The intended use should include the limitations of the app and of wanted results by the user, setting reasonable expectations for the user. As a complementary instruction, the user should be given warnings of possible effects when the app is used incorrectly or with any underlying health issue. It should be clear if the app is not suitable for users with underlying health issues or with other diagnoses. It should also state that if there are some unclearities whether the app is suitable for the user or not, the user should be advised by a doctor first.

6.1.2 Data Security and Risk of Breach

Health-related data is highly sensitive, and unauthorized access to or disclosure of such data could lead to discrimination and emotional distress for users.

Solution: Direct storage and transmission of users' real data should be avoided. Data encryption and anonymization methods should be implemented to ensure compliance with data protection regulations, and data access should be limited to authorized personnel.

6.1.3 AI Accuracy

Since the questions are generated by and a LLM they have a risk of being incorrect, outdated or misleading. Users might therefore take wrong advice or learn incorrect behaviour. Health is a living subject and some information that was correct last year might be incorrect this year.

Solution: All of the questions should be fact checked continuously and they should also be restricted to follow the guidelines of "Livsmedelsverket".

6.1.4 Manipulation

We want the user to want to come back to the app everyday. To achieve this we will implement streaks, quizzes and rewards in the app. However, streaks induce stress and could potentially give the user bad conscience. Rewards and streaks also have the possibility to generate addictive behaviour.

Solution: The streaks should be easily recovered or have different freezing options, taking "Duoling" as an example. Positive feedback is also good to have. For example, "welcome back! Do you want to start where we left off?". The user should also be able to adjust their frequency of use, with a usage alert. For example the user could be reminded every 5 minutes to take a break.

6.1.5 Bias & Inclusivity

LLMs rely on large amounts of internet-based data, which can introduce bias. For instance, much of the available information is Western centered, meaning the generated questions may unintentionally exclude or misrepresent people from other cultural or dietary traditions. Furthermore, our app may not be suitable for certain groups, such as very elderly users or those with visual impairments, which exacerbates inequalities in the digital health field.

Solution: Since food is important for some traditions the questions should not favour any specific area. They should be personalized to the user. We should create a diverse question dataset, and generate questions based on the user's habits. If he or she only eats asian food, the questions should also be based on asian foods. Furthermore, LLM should be prohibited from generating stigmatizing and judgmental language. When designing an app, the interface should be simple rather than overly complex, using clear and easy-to-understand non-technical language, and may include integrated voice prompts. Conduct usability testing for different user groups to ensure that all users have equal access to healthy learning opportunities.

6.2 Sustainability

Under construction.

7 Conclusion

Under construction.

8 Individual Contributions

8.1 Mert Affan

- Frontend implementation (Natively integration and UI design).
- Planning of meetings and overall project management.
- Writing of the Introduction, Background, and parts of the Discussions.
- Development and logic of the Quiz system flow.

8.2 Hao Tang

- Research and writing of the Background section.
- Development of the comprehensive Ethical and Sustainability Considerations section.
- Compilation of Usability Testing results.
- Definition of health intentions and user profiling criteria.

8.3 Katrine Risto

- Implementation of the Database architecture and Question Bank schema (TypeScript).
- Development of the LLM-Based Question Generation Pipeline (Prompt engineering, API integration, and Deduplication logic).
- Design and coding of the Personalized Recommendation Algorithm (Tag-based scoring and filtering).
- System architecture diagrams and technical methods documentation.

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Appendix 1: Exact instructions used for generating batches...

Exact instructions used for generating batches of questions from the different LLM´s

```
# STRICT INSTRUCTIONS FOR QUIZ QUESTION GENERATION

## SOURCE REQUIREMENTS
You must generate questions based EXCLUSIVELY on these 2 sources:
1. Livsmedelsverket.se (Swedish Food Agency)
2. https://nutritionsource.hsph.harvard.edu/ (Harvard Nutrition Source)

DO NOT use any other sources, personal knowledge, or general nutrition facts.

## QUANTITY & CATEGORIZATION
Generate 100 total questions distributed as follows:
- 40 easy questions (simple facts, basic recommendations)
- 40 medium questions (combining concepts, moderate complexity)
- 20 hard questions (nuanced topics, detailed understanding)

## HEALTH INTENTION DISTRIBUTION
Generate questions across these 5 health intentions (distribute evenly):
1. 'Improve eating pattern and daily food choices'
2. 'Reduce sugar intake'
3. 'Increase vegetable and fruit intake'
4. 'Improve protein intake'
5. 'Manage weight in a healthy way'

## STRICT OUTPUT FORMAT
You MUST return a valid JSON array containing exactly 100 bilingual question objects.

### EACH QUESTION OBJECT STRUCTURE:
{
  "english": {
    "id": "unique_id_1",
    "question_text": "string (concise, clear question in English ONLY)",
    "answer_options": ["English option 1", "English option 2", "English option 3", "English
option 4"],
    "correct_answer": 0,
    "explanation": "string (1-2 sentences of learning guidance in English ONLY)",
    "health_intention": "string (MUST be one of the 5 listed health intentions)",
    "tags": ["tag1", "tag2", "tag3"],
    "difficulty": 1
  },
  "swedish": {
    "id": "unique_id_1",
    "question_text": "string (exact Swedish translation of English question)",
```

```
   "answer_options": ["Swedish option 1", "Swedish option 2", "Swedish option 3", "Swedish option 4"],
   "correct_answer": 0,
   "explanation": "string (exact Swedish translation of English explanation)",
   "health_intention": "Improve eating pattern and daily food choices", // Keep English text
   "tags": ["tag1", "tag2", "tag3"], // Same as English
   "difficulty": 1 // Same as English
  }
}
```

DIFFICULTY LEVEL MAPPING:
- Easy: difficulty: 1 (simple recall, basic facts)
- Medium: difficulty: 2 (application, comparison)
- Hard: difficulty: 3 (analysis, synthesis, detailed concepts)

QUESTION CONTENT REQUIREMENTS

MANDATORY RULES:
1. Source fidelity: Each question must be directly supported by Livsmedelsverket.se OR Harvard Nutrition Source
2. No medical advice: Avoid diagnosis, treatment, or medical recommendations
3. Positive language: Use encouraging, educational tone
4. Actionable focus: Questions should promote practical, healthy behaviors
5. Plausible distractors: Incorrect options must be reasonable but clearly wrong
6. Language purity:
   - English version: All text in English
   - Swedish version: All text in Swedish (accurate translation)
   - Health intention field: Keep in English for both versions

QUESTION TYPES TO INCLUDE:
- Daily food choices and habits
- Nutrient-specific questions (fiber, protein, vitamins, etc.)
- Portion sizes and frequency recommendations
- Food group comparisons
- Healthy cooking and preparation methods
- Reading nutrition labels
- Hydration and beverage choices
- Meal timing and patterns

TAGS TO USE:
Include relevant tags from: whole-grains, carbohydrates, healthy-choices, fish, omega-3, recommendations, fats, heart-health, cooking, protein, plant-based, hydration, healthy-drinks, sugar, fiber, vegetables, nutrition, salt, sodium, vitamin-d, bone-health, muscle-health

TRANSLATION REQUIREMENTS
For each question:
1. Exact meaning preservation: Swedish translation must match English exactly

2. Cultural appropriateness: Use Swedish food terms familiar in Sweden
3. Metric units: Use grams, liters (not cups, pounds)
4. Swedish agency references: Use "Livsmedelsverket" not "Swedish Food Agency" in Swedish text

GENERATION PROCESS
1. Source check: Before writing each question, verify it exists in one of the two sources
2. Difficulty assignment: Consciously assign difficulty level based on complexity
3. Health intention: Explicitly connect to one of the 5 intentions
4. Translation accuracy: Create English version first, then precise Swedish translation
5. ID generation: Use unique IDs like "q001", "q002", etc.

PROHIBITED CONTENT
- Medical diagnoses or treatments
- Extreme or restrictive diet recommendations
- Weight loss supplements or products
- Non-scientific claims (superfoods, detoxes, etc.)
- Any content not found in the two specified sources
- Questions about specific health conditions
- Personal anecdotes or opinions

VALIDATION CHECKLIST FOR EACH QUESTION:
- [ ] Based on Livsmedelsverket.se OR Harvard Nutrition Source
- [ ] Positive, educational tone
- [ ] Clear single correct answer
- [ ] 3 plausible but incorrect alternatives
- [ ] 1-2 sentence explanatory answer
- [ ] One of the 5 health intentions
- [ ] Appropriate difficulty level (1, 2, or 3)
- [ ] Relevant tags included
- [ ] English version complete and accurate
- [ ] Swedish translation matches English exactly
- [ ] No medical advice or diagnoses

FINAL OUTPUT INSTRUCTIONS
You MUST return ONLY a valid JSON array with 100 objects. No markdown, no explanations, no additional text.

Example of first 2 questions in the array:

```
[
  {
    "english": {
      "id": "q001",
      "question_text": "Which type of grain is recommended as the healthiest daily choice?",
      "answer_options": ["White rice", "Whole grains", "Instant noodles", "Refined pasta"],
      "correct_answer": 1,
      "explanation": "Whole grains are recommended because they contain fiber, vitamins, and minerals that support long-term health.",
```

```
    "health_intention": "Improve eating pattern and daily food choices",
    "tags": ["whole-grains", "carbohydrates", "healthy-choices"],
    "difficulty": 1
   },
   "swedish": {
    "id": "q001",
    "question_text": "Vilken typ av spannmål rekommenderas som det hälsosammaste
dagliga valet?",
    "answer_options": ["Vitt ris", "Fullkorn", "Snabbnudlar", "Raffinerad pasta"],
    "correct_answer": 1,
    "explanation": "Fullkorn rekommenderas eftersom de innehåller fiber, vitaminer och
mineraler som stödjer långsiktig hälsa.",
    "health_intention": "Improve eating pattern and daily food choices",
    "tags": ["whole-grains", "carbohydrates", "healthy-choices"],
    "difficulty": 1
   }
  },
  {
   "english": {
    "id": "q002",
    "question_text": "How many portions of fish does Livsmedelsverket recommend per
week?",
    "answer_options": ["0 portions", "1-2 portions", "3-5 portions", "Daily"],
    "correct_answer": 1,
    "explanation": "Livsmedelsverket recommends eating fish 2-3 times per week, but at
least 1-2 portions.",
    "health_intention": "Improve eating pattern and daily food choices",
    "tags": ["fish", "omega-3", "recommendations"],
    "difficulty": 1
   },
   "swedish": {
    "id": "q002",
    "question_text": "Hur många portioner fisk rekommenderar Livsmedelsverket per
vecka?",
    "answer_options": ["0 portioner", "1-2 portioner", "3-5 portioner", "Dagligen"],
    "correct_answer": 1,
    "explanation": "Livsmedelsverket rekommenderar att äta fisk 2-3 gånger per vecka, men
minst 1-2 portioner.",
    "health_intention": "Improve eating pattern and daily food choices",
    "tags": ["fish", "omega-3", "recommendations"],
    "difficulty": 1
   }
  }
  // ... 98 more question objects exactly like this
]

## REMINDERS:
- 100 total questions (40 easy, 40 medium, 20 hard)
```

- Only 2 sources: Livsmedelsverket.se and Harvard Nutrition Source
- 5 health intentions distributed evenly
- Bilingual: Each question has identical English and Swedish versions
- Strict JSON format: No deviations from the specified structure
- Start generating now and return ONLY the JSON array